Stochastic Sampling is Epistemically Shallow: The Dimensionality Gap Between Temperature Variation and Model Diversity in LLMs

Izhar Ali [1]

Abstract

Ask a language model the same question a hundred times and the answer sometimes changes. Self-consistency reads that flicker as uncertainty about the question. We ask whether the flicker also couples questions: on a bad pass, does the model miss a cluster of related items together, the way a weak ensemble member does? On the same 500 MMLU questions we compare one model sampled 100 times at $\tau = 1$ against 24 models decoded once each at $\tau = 0$. A Marchenko–Pastur test sets a noise ceiling on both correctness matrices. Within one model the test runs on the items the model is not already certain about either way — 121 of 500 for Qwen2.5-7B, 15 to 403 across the other settings. On those items at most one eigenvalue clears the ceiling. The one case that does clear it sits inside Tracy–Widom sampling noise, and MMLU, HellaSwag, and GSM8K give the same ceiling. Across the 24 models four eigenvalues clear it, and a difficulty-matched Bernoulli null never produces more than one in 500 draws; five Qwen models of one family already produce one direction far above their own edge. The finding is scoped to the sampling protocol self-consistency actually uses, one independent draw per prompt. Under it, extra samples of one model sharpen a per-question probability. They do not recover the cross-question structure a second, independent model supplies.

1. Introduction

Sample Qwen2.5-7B a hundred times at temperature 1 on a fixed MMLU item. On most items every pass agrees: right every time, or wrong every time. On 121 of 500 items the model is right on some fraction p_i of the hundred passes. Self-consistency (Wang et al., 2023) and its relatives (Kadavath et al., 2022; Manakul et al., 2023; Farquhar et al., 2024) read that kind of variation as uncertainty about the item, and averaging independent draws is a good way to estimate one probability. We ask whether the hundred-pass record carries a second kind of information.

The second kind is covariance across items. If pass 17 is a bad pass, does Qwen miss a cluster of history items together, the way one weak member of an ensemble does? Deep ensembles (Lakshminarayanan et al., 2017) are valued for exactly that: members are systematically strong or weak over whole domains, so their disagreements point at where knowledge is missing rather than at how noisy one answer is (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Temperature rescales every logit by one constant. That leaves the noise on a chemistry item and the noise on a history item no common factor to share. A related line of work reports that LLMs do not reliably sample from a target distribution when asked to (Gu et al., 2026): a different measurement, the same suspicion that sampled variation is thinner than it looks. Whether that architectural fact shows up in the samples is an empirical question.

The measurement is a $K \times N$ matrix of 0s and 1s: K completions, N questions, and a 1 where that completion matched the gold answer. If questions flip independently, the correlation matrix of the columns is pure sampling noise, and no eigenvalue can exceed the Marchenko–Pastur edge (Marčenko & Pastur, 1967) by more than Tracy–Widom fluctuations. An eigenvalue above that edge means questions move together. We run the same test on a second matrix whose rows are 24 different models, each decoded once at $\tau = 0$. That ensemble is a calibration reference rather than a competitor: it shows what the statistic recovers when a shared factor is present, so a within-model null reads as an absence of signal and not as a failure of the test.

Figure 1 shows the result: noise on one side, four directions above the edge on the other.

Contributions. The first measurement is the within-model ceiling. Across five instruction-tuned models, three bench-

[1]Rowan University. Correspondence to: Izhar Ali <aliizh94@rowan.edu>.

2nd Workshop on Epistemic Intelligence in Machine Learning (EIML@ICML 2026), Seoul, South Korea. *Copyright 2026 by the author(s).*

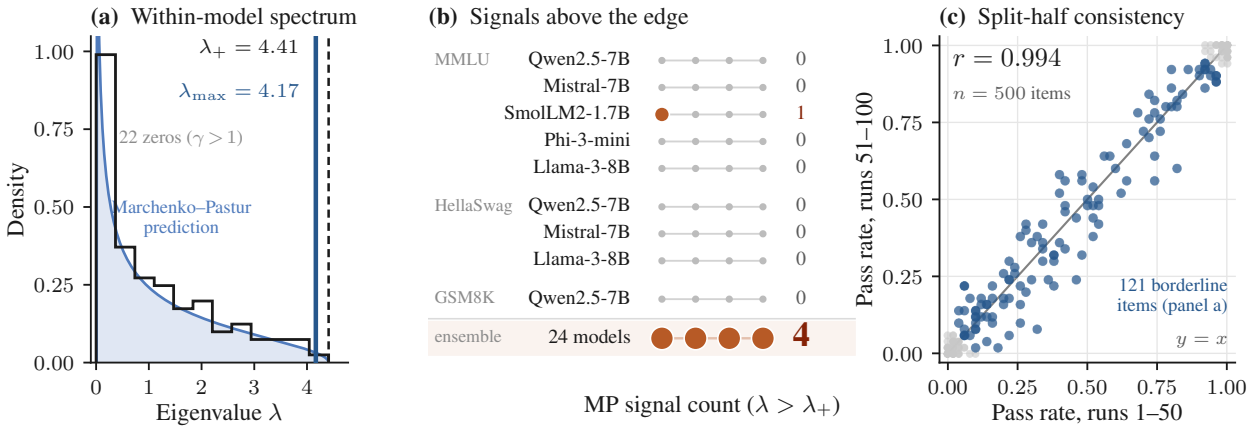


Figure 1. **The dimensionality gap. (a)** Eigenvalue density of Qwen2.5-7B's question–question correlation matrix ($K = 100$, $N_b = 121$) against the Marchenko–Pastur law. The bulk matches the prediction; $\lambda_{\max} = 4.17$ sits below $\lambda_+ = 4.41$. **(b)** MP signal count (eigenvalues above λ_+) for every within-model setting in Table 1, against the 24-model ensemble. At most one versus four. **(c)** Split-half pass rates on the same 500 items ($r = 0.994$): repeated samples recover each p_i.

marks, and the full range of the cut that selects which items the test sees, at most one eigenvalue of the correctness correlation matrix clears the MP edge. The single case that does clear it stands $0.05\,\sigma_{\mathrm{TW}}$ above it. The second measurement is the contrast. The same test on 24 models returns four directions, and a difficulty-matched Bernoulli null returns at most one on every one of 500 draws. Curated cross-architecture diversity is not obviously what drives this: five Qwen models at four different scales, one family, already give a leading eigenvalue of 129.9 against an edge of 68.1.

Related work. Across models this kind of structure is already documented. A single factor captures 79% of variance on 5,000+ leaderboard models (Kipnis et al., 2025), factor analysis of 60 models yields eight dimensions (Maimon et al., 2025), and joint embeddings place models in a low-dimensional space (Yao et al., 2025). All three analyze a model-by-benchmark score matrix. The within-model object — runs by questions, read against a sampling-noise null — has not been measured. Hamidieh et al. (2025) empirically showed that cross-model disagreement captures uncertainty self-consistency misses. We ask whether that disagreement is a shared factor over questions, not only a better score. Kim et al. (2025) measured pairwise error correlation across 350+ models, one number per pair rather than the spectrum, and Wen et al. (2025) factor-analyzed within-question answer-choice variation rather than the cross-question covariance studied here.

2. Method

One model and N questions yield K completions of each question at temperature $\tau > 0$, each from a fresh forward pass with its own sampling RNG and no shared KV cache. Entry c_{ki} is 1 when run k answers question i correctly. The matrix $\mathbf{C}$ is $K \times N$, and every entry is scored against the gold answer. That is deliberate. The question is structural — does the sampling process couple questions — and covariance in correctness is what that question is about, so labels are needed to ask it at all. Self-consistency's deployable confidence is a different statistic, the fraction of the K samples agreeing with the modal answer, which needs no labels. On this data the two should track each other closely, because Qwen's errors on a borderline item — one it gets right on some passes and wrong on others — concentrate on one wrong letter (§3.1). Closely is not identically, and nothing below assumes the two are one quantity.

A row of $\mathbf{C}$ is not a trajectory that carries a latent state across the benchmark. It is a slice of N independent per-prompt draws, which is how self-consistency is run in practice (Wang et al., 2023). Assumption A.1 states that independence formally; under it, any correlation the test finds would have to come from the model rather than from sampling noise. The assumption is operational and we did not verify it directly: shared RNG state, batching, or tokenization could couple prompts in a way this test would report as model structure.

Questions that are always right or always wrong have no variance and cannot inform a correlation test. We call a question *borderline* when its pass rate $f_i = \frac{1}{K}\sum_k c_{ki}$ lies in $(\varepsilon, 1-\varepsilon)$ with $\varepsilon = 0.05$, and we keep those N_b columns. Every within-model result below is a statement about the borderline items, not about all 500.

We treat the borderline questions as variables and the K runs as observations, and let $\mathbf{R}$ be their Pearson correlation matrix. Independent questions still look weakly correlated in a finite sample. The Marchenko–Pastur edge (Marčenko & Pastur, 1967) $\lambda_+ = (1 + \sqrt{\gamma})^2$, $\gamma := N_b/K$, is the height those accidental correlations are allowed to reach. Anything taller is structure. The standardized top eigenvalue $z := (\lambda_1 - \lambda_+)/\sigma_{\mathrm{TW}}$ converges to Tracy–Widom F_1 (Prop. A.2) and is the test. The count of eigenvalues above λ_+ is a summary, not a calibrated p-value, and Horn's parallel analysis (Horn, 1965) (200 column-shuffle surrogates) is the cross-check.

The across-model matrix replaces the K stochastic runs with M models, each decoded once at $\tau = 0$, and keeps only questions on which the models split. The test itself does not change. Its null is 500 independent-Bernoulli matrices whose column rates are fixed at the observed p_i. That preserves how hard each question is and destroys model identity.

We sample five instruction-tuned models (Qwen2.5-7B, Mistral-7B-v0.3, SmolLM2-1.7B, Phi-3-mini, Llama-3-8B) at $K = 100$, $\tau = 1$, on 500 MMLU test items (Hendrycks et al., 2021). The same test is repeated on HellaSwag (Zellers et al., 2019) (three models, $K = 50$, $N = 200$) and on the first 100 GSM8K test problems (Cobbe et al., 2021) (Qwen, $K = 30$). The across-model matrix is one greedy pass of 24 models, 0.5B to 22B, on the same 500 items. Twenty-three are instruction-tuned and Phi-2 is a base model; the roster and protocol are in App. B.

3. Results

3.1. One model gives at most one direction

Qwen2.5-7B is right on 71.2% of the 500 items at $\tau = 1$ (77.4% greedy), and 121 of them are borderline. On those 121 the spectrum of $\mathbf{R}$ matches the Marchenko–Pastur prediction in Fig. 1a, and the top eigenvalue is 4.17, below $\lambda_+ = 4.41$. Being right on a history item on a given pass does not predict a chemistry item on the same pass, or even another history item: mean correlation among borderline items in the same MMLU subject is -0.003, and -0.0002 across subjects.

The same ceiling holds in every row of Table 1 and every within-model bar in Fig. 1b. Mistral, Phi-3-mini, and Llama-3-8B sit below the edge. SmolLM2 touches it ($\lambda_{\max} = 9.05$ against $\lambda_+ = 9.04$, TW $z = +0.05$): one eigenvalue above the raw edge, zero by Horn. That is ordinary under a finite-K null whose mean z at $K = 100$ is about -1.60. Every observed z in the table lies in the central 90% of that null. SmolLM2 also stays on the signal side of the edge across the whole ε sweep, reaching $z = +1.77$ at $\varepsilon = 0.15$, near the 99th percentile of the null: slight residual coupling that Assumption A.1 idealizes away, with chat-template randomness a candidate mechanism. Its raw count touches two at one setting in the sweep ($\varepsilon = 0.10$), while the buffered count stays at zero throughout — the raw count's known noise sensitivity (Prop. A.2), not a second real direction.

Table 1. Marchenko–Pastur test. Positive margin: $\lambda_{\max}$ below the noise edge. SmolLM2 is the one case with a count of 1 (no buffer on λ_+). TW z is distance from the edge in units of σ_{TW}; at $K = 100$ the null mean is about -1.60.

Model	N_b	λ_+	$\lambda_{\max}$	**margin**	**TW** z
MMLU ($K = 100$, $N = 500$)					
Qwen2.5-7B	121	4.41	4.17	$+0.24$	-2.01
Mistral-7B-v0.3	283	7.19	6.85	$+0.34$	-2.37
SmolLM2-1.7B	403	9.04	9.05	-0.01	$+0.05$
Phi-3-mini	269	6.97	6.78	$+0.19$	-1.36
Meta-Llama-3-8B	173	5.36	5.05	$+0.31$	-2.37
HellaSwag ($K = 50$, $N = 200$)					
Qwen2.5-7B	15	2.40	2.21	$+0.19$	-1.15
Mistral-7B-v0.3	58	4.31	4.04	$+0.28$	-1.45
Meta-Llama-3-8B	82	5.20	4.80	$+0.40$	-1.96
GSM8K ($K = 30$, $N = 100$)					
Qwen2.5-7B	65	6.11	5.72	$+0.39$	-1.27

The ceiling is not an MMLU artifact, since HellaSwag and GSM8K give it too, GSM8K on a multi-step chain-of-thought trace. It is not an artifact of the borderline cut either (Fig. 2a). Scoring only correct against incorrect could in principle hide a shared wrong answer. When Qwen errs on a borderline item it picks the same wrong letter 87.4% of the time, and those local preferences still do not associate across items (App. E). That appendix also records that the letter-level run is a second $K = 100$ pass, not the matrix behind Table 1.

3.2. Twenty-four models give four

The same test on 24 models finds structure. Four eigenvalues sit above the MP edge on the 459 questions where at least one model disagrees (Fig. 1b): $\lambda_{1..4} = 83.0$, 43.1, 36.5, and 34.6 against $\lambda_+ = 28.9$. A difficulty-matched Bernoulli null — independent coin flips at the observed per-question rates — produces at most one signal eigenvalue on every one of 500 draws (mean 0.028), never four.

Family membership does not obviously account for the four. Same-family pairs correlate a little more than cross-family pairs ($\bar{r} = 0.41$ against 0.34, Cohen's $d = 0.58$), and the two densities overlap for most of the 276 pairs (Fig. 2b; same pattern in a ranking task, App. D). Single-family sub-ensembles also clear the edge on their own: five Qwen models (the four Qwen2.5 sizes plus Qwen2-7B) give a leading eigenvalue of 129.9 against $\lambda_+ = 68.1$ on 263 split

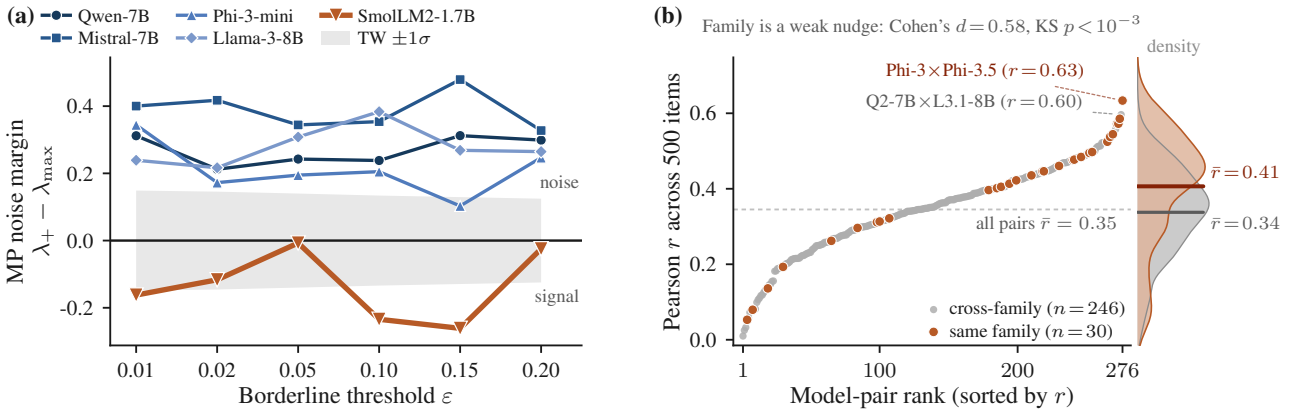


Figure 2. **Neither the borderline cut nor family membership explains the gap. (a)** MP noise margin $\lambda_+ - \lambda_{\max}$ against the borderline threshold ε for the five MMLU models ($K = 100$). Positive margin: no eigenvalue past the edge. Four models stay positive; SmolLM2 stays slightly negative. Grey band: Tracy–Widom $\pm 1\sigma$. **(b)** All 276 pairs among the 24 models, sorted by Pearson r across the 500 items. Red: same family ($n = 30$). Grey: cross-family ($n = 246$). Family is a weak nudge (Cohen's $d = 0.58$); the densities overlap for most pairs.

questions, and four Llama models give 208.9 against 100.8. Each sub-ensemble returns a count of one — an empirical result, not a constraint: the rank of a five-model and a four-model correlation matrix would permit four and three. Of those nine models, only Qwen2.5-7B and Meta-Llama-3-8B were also probed alone at $K = 100$, and neither showed a direction above its own edge (Table 1). The other seven enter only as single greedy passes, so no within-model test exists for them. None of this identifies the four directions: we did not load the eigenvectors onto families, so whether they are lineage, capability, or a mixture is open.

4. Discussion

The samples carry no shared factor because the protocol supplies none. Each prompt is an independent draw, and temperature does not single out any one topic. Model identity is such a factor, which is why the same test returns four directions across 24 models and one direction inside a single family sampled at four scales. The scope follows from the mechanism, and it is narrower than the title compresses: what we measured is sampling under the independent-draw protocol self-consistency actually uses. A protocol that installed a shared latent — a fixed prompt perturbation, a common dropout mask (Gal & Ghahramani, 2016), a benchmark-wide logit bias — could produce cross-question structure out of one model. That experiment separates a shallow model from a shallow protocol. We did not run it.

Four is a count taken at $M = 24$, and it can mislead in either direction. It may undercount, for two reasons that are not the same reason. We scored correctness only, so any structure carried in log-probabilities, hidden states, or answer identity is invisible to this test. And the measurement has not saturated: effective rank over subsampled ensembles climbs with M all the way out to 24 (median $\exp(H)$ of 2.89 at $M = 4$, 12.8 at $M = 16$, 18.7 at $M = 24$), tracking the $M - 1$ ceiling with no flattening. Effective rank tracks that ceiling under the null too (App. C). So the sweep shows only that the measurement is still bounded by ensemble size, not that a larger pool would add real directions. Separately, four may overcount what it is taken to mean. Models that share pretraining corpora can be wrong on the same items for the same reason, and a direction built out of shared training data is not complementary epistemic coverage. Undercounting the rank and overcounting the coverage are different risks, and neither is settled here.

Two further limits bound the within-model side. The test runs at $\gamma = N_b/K$ between 0.3 and 4.0; $K \gg N_b$ would sharpen detection of a direction at SmolLM2's edge. All five probed models are instruction-tuned, so we cannot say whether a pretrained-only base model behaves the same way. An open-generation benchmark, a 70B-tier probe, and a log-probability null are the natural next tests.

A hundred samples recover each p_i almost exactly (split-half $r = 0.994$, Fig. 1c), and that is the quantity self-consistency approximates without labels. Across questions they recover nothing. The sampling loop estimates a scalar per item without inducing a factor over items. What follows is a budgeting rule: to sharpen the confidence on one question, keep sampling; to find out which questions a model is wrong about together, more samples do not get there; one more model does.

References

Bao, Z., Pan, G., and Zhou, W. Tracy–Widom law for the extreme eigenvalues of sample correlation matrices. *Electronic Journal of Probability*, 17:1–32, 2012.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Prafulla, D., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Farquhar, S., Kuhn, J., Gal, Y., et al. Detecting hallucinations in large language models using semantic entropy. *Nature*, 2024.

Gal, Y. and Ghahramani, Z. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.

Gu, X., De, S., Titsias, M., Markeeva, L., Veličković, P., and Pascanu, R. The illusion of stochasticity in LLMs. *arXiv preprint arXiv:2604.06543*, 2026.

Hamidieh, K., Thost, V., Gerych, W., Yurochkin, M., and Ghassemi, M. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *NeurIPS Workshop on Reliable and Responsible Foundation Models*, 2025.

Hendrycks, D., Burns, C., Basart, S., et al. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.

Horn, J. L. A rationale and test for the number of factors in factor analysis. *Psychometrika*, 30(2):179–185, 1965.

Hüllermeier, E. and Waegeman, W. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110:457–506, 2021.

Kadavath, S., Conerly, T., Askell, A., et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Kendall, A. and Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Kim, E., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2506.07962.

Kipnis, A., Voudouris, K., Schulze Buschoff, L. M., and Schulz, E. metabench – a sparse benchmark of reasoning and knowledge in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.

Lakshminarayanan, B., Pritzel, A., and Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Maimon, A., Cohen, A. D. N., Vishne, G., Ravfogel, S., and Tsarfaty, R. IQ test for LLMs: An evaluation framework for uncovering core skills in LLMs. *arXiv preprint arXiv:2507.20208*, 2025.

Manakul, P., Liusie, A., and Gales, M. J. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Marčenko, V. A. and Pastur, L. A. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457, 1967.

Pillai, N. S. and Yin, J. Edge universality of correlation matrices. *Annals of Statistics*, 40(3):1737–1763, 2012. arXiv:1112.2381.

Wang, X., Wei, J., Schuurmans, D., et al. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Wen, Z., Liu, Z., Tian, Z., Pan, S., Huang, Z., Li, D., and Huang, M. Scenario-independent uncertainty estimation for LLM-based question answering via factor analysis. In *Proceedings of the ACM Web Conference*, 2025.

Yao, L. H., Jarvis, N., Zhan, T., Ghosh, S., Liu, L., and Jiang, T. JE-IRT: A geometric lens on LLM abilities through joint embedding item response theory. *arXiv preprint arXiv:2509.22888*, 2025.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a machine really finish your sentence? In *Association for Computational Linguistics (ACL)*, 2019.

A. Null Prediction and Test Statistic

Assumption A.1 (Per-question independence). Assume $c_{ki} \sim \mathrm{Ber}(p_i)$ are mutually independent across questions i within a run and across runs k. Operationally: independent forward passes with a fresh sampling RNG per question and no shared KV cache.

Proposition A.2 (Null spectrum and test statistic)**.** *Under Assumption A.1 with $p_i \in [\varepsilon + \delta, 1 - \varepsilon - \delta]$ for all borderline questions, some fixed $\delta > 0$: (i) the population Pearson correlation matrix of $\mathbf{C}_B$'s columns is $\mathbf{I}_{N_b}$; (ii) as $K \to \infty$ with γ fixed, $\mathbf{R}$'s empirical spectrum converges to Marchenko–Pastur on $[\lambda_-, \lambda_+]$ and the standardized top eigenvalue $z = (\lambda_1 - \lambda_+)/\sigma_{\mathrm{TW}}$ converges in distribution to Tracy–Widom F_1, with edge scale $\sigma_{\mathrm{TW}} = (1 + \sqrt{\gamma})(1 + 1/\sqrt{\gamma})^{1/3} K^{-2/3}$. The unbuffered count $|\{i : \lambda_i > \lambda_+\}|$ does not vanish as $K \to \infty$, since $\mathbb{P}(\mathrm{TW}_1 > 0)$ is bounded away from zero, so we read it only as a summary and use z as the calibrated test.*

Sketch. Independence gives $\mathrm{Cov}(c_{ki}, c_{kj}) = 0$ for $i \neq j$ and $p_i(1 - p_i) > 0$ on borderline questions, so the population Pearson correlation is $\mathbf{I}_{N_b}$. By Hoeffding with a union bound over the N candidate questions, $\sup_i |f_i - p_i| \to 0$ in probability as $K \to \infty$, so the δ-buffer forces the data-driven borderline set $\{i : f_i \in (\varepsilon, 1 - \varepsilon)\}$ to coincide with its population counterpart with probability $\to 1$ (at $p_i = \varepsilon$ exactly, this convergence would fail: f_i lands on either side of ε with probability $\to 1/2$). For (ii), center and rescale each column by $1/\sqrt{p_i(1 - p_i)}$ (sample Pearson correlation is invariant to both) to obtain independent, bounded, unit-variance entries, not identically distributed across columns.

Pillai & Yin (2012) and Bao et al. (2012) establish Tracy–Widom edge universality for exactly this sample-correlation-matrix setting; bulk convergence to Marchenko–Pastur extends the classical i.i.d. case (Marčenko & Pastur, 1967) under the same weaker condition. At the $K = 100$ used here the finite-sample mean of z is about -1.60 rather than the F_1 mean ≈ -1.21 — a finite-K standardization shift we also measure under Gaussian and homogeneous-Bernoulli nulls, so it is not particular to unequal p_i — and we therefore read Table 1 against that Monte Carlo null, which re-applies the borderline filter inside every draw to match the selection effect it corrects for. □

B. Models and Decoding

MMLU. $N = 500$ questions from the test split, ten items from each of the first 50 subjects in alphabetical order. Seven subjects are absent: professional_psychology, public_relations, security_studies, sociology, us_foreign_policy, virology, world_religions.

Within-model models. Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, SmolLM2-1.7B-Instruct, Phi-3-mini-4k-Instruct, Meta-Llama-3-8B-Instruct. A temperature check on Qwen2.5-7B uses $\tau \in \{0.5, 1.5\}$, $K = 30$, $N = 100$. At $\tau = 1.5$, $N_b = 16$, $\lambda_+ = 2.99$, $\lambda_{\max} = 2.74$: margin $+0.25$, MP count still zero. At $\tau = 0.5$ only three questions are borderline, below the test's floor.

HellaSwag and GSM8K. HellaSwag: $N = 200$ stratified validation items, $K = 50$. GSM8K: first 100 test problems, $K = 30$, chain of thought. GSM8K accuracy on this slice is 46.5%.

Across-model roster. The same 500 MMLU questions, one greedy pass each, 24 models from 0.5B to 22B: five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B, 7B, and Qwen2-7B), five Mistral-family (v0.2, v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), four Llama, two Gemma-2, and one each of SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-7B, and Granite-3.1-8B. Family labels follow this grouping (Zephyr with Mistral), so there are ten families. Model accuracies run from 40.2% (Llama-3.2-1B) to 77.4% (Qwen2.5-7B and Llama-3.1-8B). Mean pairwise Pearson r is 0.35 (range 0.01–0.63). Same-family pairs run higher ($\bar{r} = 0.41$) than cross-family pairs (0.34).

Decoding. All stochastic runs use $\tau = 1.0$ and top-$p = 1.0$. HuggingFace `generate` is left at its default top-$k = 50$. Qwen2.5-7B and Mistral-7B were decoded from a raw multiple-choice prompt with `max_new_tokens` $= 4$. SmolLM2, Phi-3-mini, Llama-3-8B, and the 24-model ensemble used each model's chat template with `max_new_tokens` $= 6$. GSM8K uses a chain-of-thought prompt and 256 new tokens. Completions are scored by letter match or normalized text match on MMLU and HellaSwag, and by numerical extraction on GSM8K. Qwen and Mistral therefore used a slightly different prompt and length than the later within-model runs.

C. Parallel Analysis and Shannon Rank

Horn's parallel analysis (Horn, 1965) is the cross-check. We generate 200 surrogate $K \times N_b$ binary matrices by shuffling each column of $\mathbf{C}_B$ independently, which preserves per-question pass rates and destroys cross-question correlation. For each rank j we compare $\lambda_j(\mathbf{R})$ to the 95th percentile of λ_j across surrogates. The PA count is the largest r such that λ_j exceeds its surrogate threshold for every $j \le r$ (the classical contiguous rule). Counting every rank above the 95th percentile, including in the bulk, over-counts by about 5% of ranks under the null. Under the contiguous rule, PA reports 0 on every model and benchmark in Table 1, including SmolLM2, where the unbuffered MP count is 1. Both are consistent with the $d_{\text{within}} \le 1$ ceiling. SmolLM2's count is also a convention artifact: columns are centered on their own sample mean, so $\gamma = N_b/(K-1)$ is at least as defensible as N_b/K, and it moves λ_+ to 9.106 against $\lambda_{\max} = 9.05$ — a margin of $+0.05$ and a raw count of 0. A -0.01 margin on an edge near 9 sits below the resolution that convention alone can settle.

Shannon effective rank does not separate the two regimes. Let $\hat{\lambda}_j = \lambda_j / \sum_\ell \lambda_\ell$ be the correlation matrix's normalized eigenvalues and $\exp(H)$, $H = -\sum_j \hat{\lambda}_j \log \hat{\lambda}_j$, its effective rank. Under any heterogeneous-rate Bernoulli null, $\exp(H)$ saturates near the $M - 1$ ceiling (centered columns lie in the hyperplane orthogonal to $\mathbf{1}_M$). Against its own null, the observed across-model value, 18.7, sits below the matched-difficulty mean of 22.3 (min 22.1) at $M = 24$ — a real gap. Against the within-model side, the two regimes do not separate: the five MMLU settings already reach 64–87% of the $K - 1$ ceiling, next to the across-model 81%. Entropy rank detects the across-model null on its own terms and still occupies the same ceiling-adjacent band on both sides of the comparison that MP counting separates.

Subsampling the 24-model pool at $M \in \{4, 6, \ldots, 24\}$ (500 random subsets per M; at $M = 24$ there is only the one) gives a median $\exp(H)$ that falls from 96% of the $M - 1$ ceiling at $M = 4$ to 81% at $M = 24$, monotonically, with no flattening. Because the null saturates near that same ceiling, this establishes only that the across-model measurement is still limited by ensemble size at $M = 24$, which is the basis for treating four as a value at this pool size rather than as an asymptote.

D. Ranking Diagnostic

This appendix records a gold-using ranking of whether Qwen2.5-7B-Instruct is correct at $\tau = 0$ on the 500 MMLU questions (base accuracy 0.774). Both signals below are computed from gold correctness. They diagnose information already in the correctness matrices; they are not a label-free production comparison.

Sample consistency SC_K. From Qwen's K stochastic passes at $\tau = 1$, let p_i be the pass rate on question i. Confidence is $\max(p_i, 1 - p_i)$. This uses gold to score each pass. It is not modal letter agreement. The oracle pass rate p_i itself reaches AUROC 0.956 at $K = 100$; that is an upper bound on any sample-based ranker that is allowed to see labels, not a competitor. Sample consistency improves slowly with K ($+0.20$ AUROC from $K = 2$ to $K = 100$) and flattens: $K = 50$ to $K = 100$ adds 0.011.

Peer correctness Ens_k. For k other models run once at $\tau = 0$, confidence is the fraction of those models that are correct on the item. This uses gold on the peers. It is not the fraction whose answer string matches Qwen's greedy answer (peer letters were not saved). Family-diverse selection walks down $\tau = 0$ MMLU accuracy, one model per family under the ten-family grouping above. At $k \leq 3$ that rule and a pure top-k pick the same models (Llama-3.1-8B, Gemma-2-9B, Phi-3-mini).

Headline comparison. Sample consistency at $K = 100$ reaches AUROC 0.712. Two peers reach 0.807 (bootstrap $p = 0.006$ against 0.712). A single peer reaches 0.748, which is not distinguishable from 0.712 at $N = 500$ (bootstrap $p \approx 0.31$). Three peers reach 0.839. Family diversity is not what drives this. Three small Qwen2.5 models (0.5B, 1.5B, 3B), one family, already reach 0.811, against 0.839 for a family-diverse three; that gap of 0.028 was not tested for significance, and it is small next to the gap between either triple and sample consistency. Adding independent models is what moves the ranking, and family diversity is at most a second-order effect at these k. Cost in Table A1 is a parameter-FLOP proxy (parameters/7B) and ignores sequence length.

E. Local Preferences

On a second Qwen2.5-7B run of $K = 100$ with answer letters saved, 114 items are borderline. That run is not identical to the Table 1 correctness matrix (about 10% of entries differ; per-question pass rates still correlate at $r = 0.94$), so the 87.4% figure quoted in the main text is scoped to this run. When Qwen errs on those 114 items, the most common wrong letter accounts for 87.4% of the errors. A permutation test that shuffles each question's letter sequence independently finds no pairwise Cramér's V or mutual information above the 99th percentile of the null's maximum over all pairs (200 permutations) — the family-wise test, not a per-pair threshold that a few hundred pairs would clear by chance alone.

Table A1. Gold-using rankers of Qwen2.5-7B $\tau = 0$ correctness on 500 MMLU questions. Sample consistency: $\max(p, 1 - p)$ from K stochastic Qwen passes. Peer correctness: fraction of k other models that are correct. Cost: parameters/7B. Family-diverse $k = 1$ is Llama-3.1-8B.

Predictor	K/k	**AUROC**	**AUPR**	**Cost**
SC_K	100	0.712	0.880	100.00
Peer, family-diverse	1	0.748	0.874	1.14
Peer, family-diverse	2	0.807	0.905	2.46
Peer, family-diverse	3	0.839	0.924	3.00
Peer, same-family Qwen	3	0.811	0.909	0.71

The other four models have no saved letters. The same permutation test on their binary correctness patterns (1,128 additional borderline items) is also below the 99th percentile of that null maximum. A null on correctness patterns is required for a letter-level claim and does not establish one. We do not claim a five-model answer-identity result.

A one-hot MP test on Qwen's letters is invalid as an identity check: the four indicators of one question are mutually exclusive, so their population correlation is block-diagonal rather than the identity the MP edge assumes, and that mismatch alone produces three eigenvalues above $1.05\lambda_+$.